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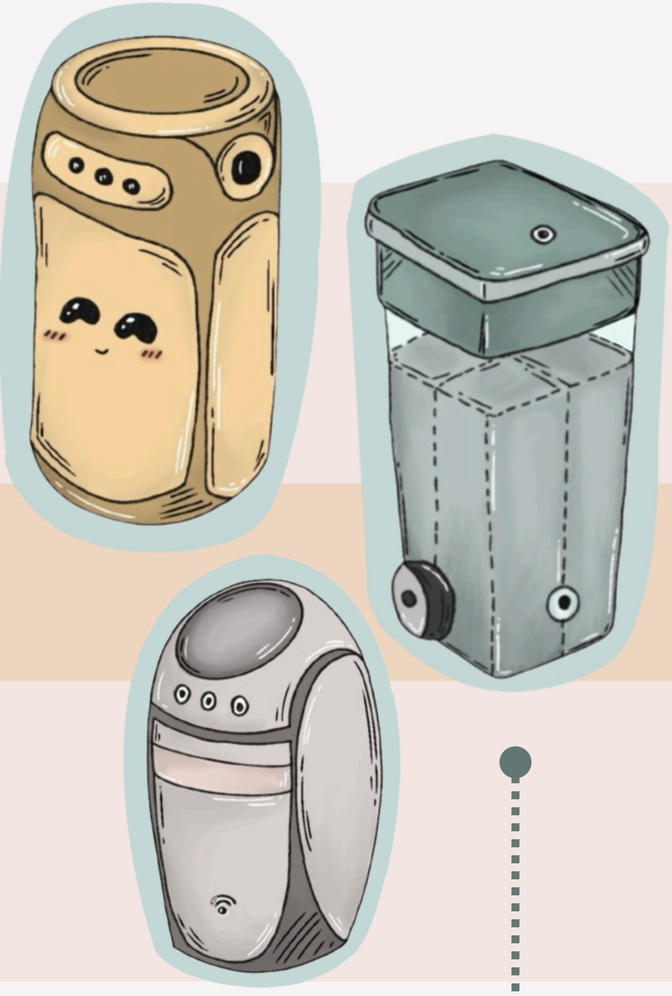
Design Process

Problem Statement

How might we create an effective cleaning solution to improve hygiene standards and efficiency for cleaning staff?



1 Initial Concept Sketch

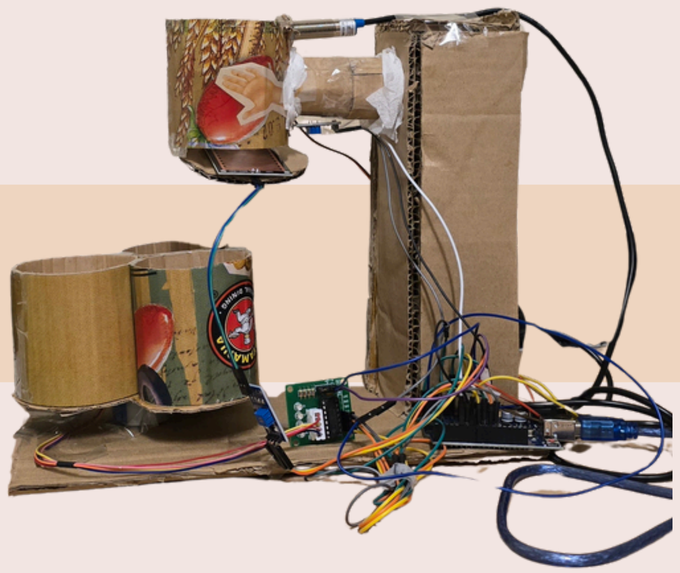
This initial sketch outlines the basic features and structure envisioned for the sorting bin.

2 First Physical Iteration

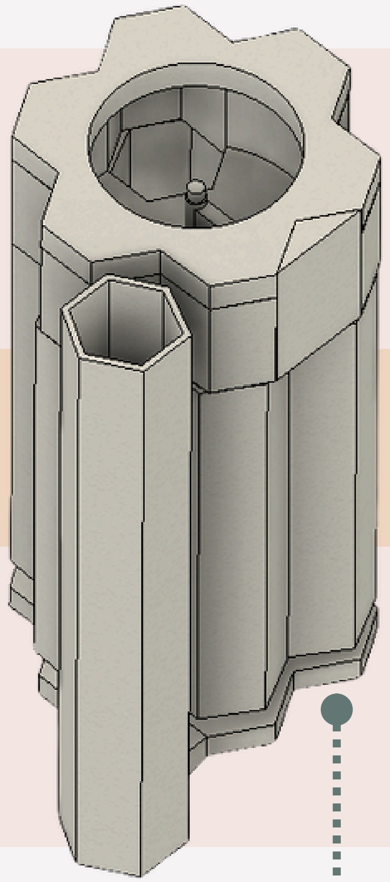
Main Goal: Ensure rotation of the motors of the spinning parts are functional.

Identified Issues:

-Difficulty in rotation of the Server Motor.
-Proximity Sensor malfunctions.
-Modular compartments constantly topples.
-IR Sensor detects bin material



Each item was tested three times to ensure consistency in sorting performance. The first prototype, built with an IR sensor and basic sorting logic, showed limited success. It achieved 66% accuracy for larger plastic, 33% for smaller plastic, and 0% accuracy for both tissue and battery waste. These shortcomings were primarily due to sensor limitations, misalignment in the rotating lid, and overly simple detection logic, which led to misclassification or failure to detect certain materials.



3 First CAD ver. Model

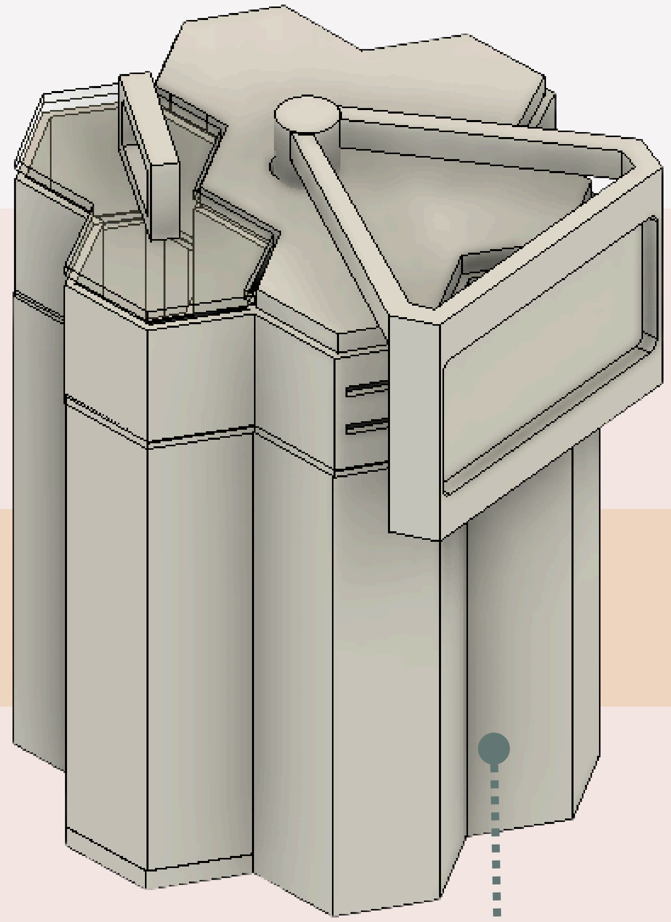
- Designed for automatic sorting after all trash is inserted
- Used hexagonal bins for a clean, modular design when detached
- Added interactive elements to enhance user engagement

4 Developed Physical Prototype

- Switched to sorting items one by one for better control and accuracy
- Replaced traditional sensors with ESP32-CAM module for better detection accuracy
- Added a transparent top cover to allow natural light into the camera module

Type of trash	Size (cm)	First Prototype Sorting Accuracy	Final Product Sorting Accuracy
Plastic	4.5 x 3	66%	**70 %**
	3 x 3	33 %	**50%**
tissue	5.5 x 4.5	0	**50%**
	4.5 x 3	0	**50%**
Battery	4.8 x 2.5	0	**50%**

Testing Accuracy Comparison: Prototype vs Final Product



5 Finalized CAD

- Extended screen display for easier interaction.
- Reusing recycled material instead of PLA

In contrast, the final prototype demonstrated 50% sorting accuracy across all tested categories, including plastic, tissue, and battery. This significant improvement was achieved by implementing several key upgrades: better mechanical alignment, replacement of the IR sensor with an ESP32-CAM module, and refined sorting algorithms. These enhancements allowed the system to reliably detect and correctly sort various waste types, making it a dependable and functional automatic trash-sorting bin.